Lesson 6.1 Guided Notes

Session 1 · Dividing by a Unit Fraction

I can divide a whole number by a unit fraction and explain what the quotient counts. · 6.NS.A.1

Word / Spanish	Meaning — complete the blank	Example
unit fraction / fracción unitaria	A fraction with _____ as its numerator.	1/3, 1/8, 1/10
reciprocal / recíproco	The number you multiply by to get a product of _____.	1/4 and 4: 1/4 × 4 = 1
quotient / cociente	The _____ to a division problem.	In 8 ÷ 1/2 = 16, 16 is the quotient

Key idea

Dividing by a unit fraction asks how many _____ of that size fit into the number.

Dividing by 1/4 is the same as multiplying by _____, so 3 ÷ 1/4 = 3 × 4 = 12.

The quotient is _____ than the whole number you started with.

Worked example

How many 1/4-pound balls of clay can you make from 3 pounds of clay?

Each pound holds _____ fourths.

3 pounds hold 3 × 4 = _____ fourths.

Check: 12 × 1/4 = _____ pounds.

Answer: 3 ÷ 1/4 = _____ balls of clay

You try

1. Divide.
a. 6 ÷ 1/2 = _____
b. 5 ÷ 1/5 = _____

_____

2. A 2-yard ribbon is cut into 1/8-yard pieces. Use the bars to find how many pieces.



_____

Sum it up: Dividing a whole number by a unit fraction gives a greater number because ___.

_____

Session 2 · Dividing a Fraction by a Whole Number

I can divide a fraction by a whole number and explain the result. · 6.NS.A.1

Word / Spanish	Meaning — complete the blank	Example
reciprocal / recíproco	Two numbers are reciprocals when their product is _____.	8 and 1/8: 8 × 1/8 = 1
multiplicative inverse / inverso multiplicativo	Another name for the _____.	2/3 and 3/2
divisor / divisor	The number you _____ by.	In 3/4 ÷ 2, the divisor is 2

Key idea

Dividing a fraction by a whole number _____ it into equal parts.

To divide by 4, multiply by its reciprocal, _____.

Each share is _____ than the fraction you started with.

Worked example

Share 2/5 pound of rice equally among 4 people.

Write 4 as a fraction: 2/5 ÷ _____.

Multiply by the reciprocal: 2/5 × _____ = 2/20.

Simplify: 2/20 = _____.

Answer: Each person gets _____ pound of rice.

You try

1. Divide. Write each quotient in simplest form.
a. 1/3 ÷ 4 = _____
b. 4/7 ÷ 2 = _____

_____

2. 5 friends share 5/8 of a pizza equally. What fraction of the whole pizza does each friend get?

_____

Sum it up: When I divide a fraction by a whole number, each share is ___ because ___.

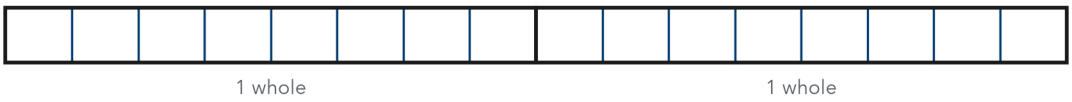
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Session 1 · Model, reason, and explain

Dividing by a Unit Fraction · 6.NS.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

How many 1/3-meter strips fit in 4 meters? Draw a measurement model and explain what the quotient counts.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: assumes division by a fraction must reduce the number.. Explain what changes and what stays true.

_____

_____

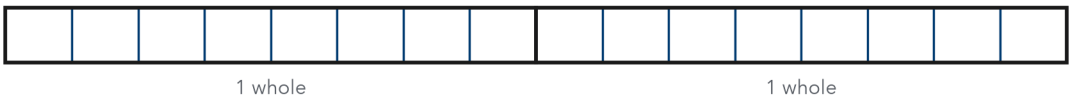
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Dividing a Fraction by a Whole Number · 6.NS.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Share 3/4 meter of ribbon equally among 3 groups. Draw the sharing, then connect it to multiplication by the reciprocal.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: confuses number of groups with size of a group.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?